

Lab 3: Architectural Pattern Justification

System: Digital Gift Card & Loyalty Point Exchange

Name: Prathamesh Kumar Singh

SRN:PES1UG24CS341

Architecture Selection: We chose the **Microservices Architecture** for the Digital Gift Card & Loyalty Point Exchange System.

Reason 1 — Independent Scaling for Uneven Load

The platform aggregates loyalty points from many Merchant Partners, converts them at dynamic rates, and redeems gift cards. Merchant integration traffic, point-conversion requests, and voucher redemption spikes (e.g., during promotions) all follow different, uncorrelated load patterns. A microservices split lets the Merchant Integration Service, Point Conversion Service, and Voucher Generation Service each scale independently, rather than forcing the whole system to scale as one monolithic unit.

Reason 2 — Fault Isolation Around a Financial Ledger

Because point conversion and voucher redemption directly affect the user's real balance, a failure or slowdown in one merchant's integration (e.g., a Merchant Partner API outage) must not bring down wallet balances or gift-card redemption for the rest of the platform. Deploying each capability as an independent service confines failures to that service, which is critical for a financial exchange system with a zero-ledger-drift requirement.

Security Advantage

Isolating the Fraud Detection & Voucher Security Service as its own microservice lets it enforce anti-fraud voucher locking and cryptographic checksum validation as a single, independently hardened trust boundary. Every point-conversion and voucher-generation request must pass through this service's provided interfaces (FraudCheck, VoucherValidation), so access controls, audit logging, and rate limiting can be tightened in one place without touching business logic elsewhere, and a compromise of a less-sensitive service (such as Merchant Integration) does not directly expose fraud-control internals.

Performance Benefit

Splitting the system into focused services allows the Point Conversion Service and Fraud Detection Service to be deployed close together with dedicated, right-sized resources (e.g., in-memory rate caches, optimized database indexes), which helps the system meet the NFR-001 target of processing point conversion and gift card redemption within 250 ms, since these hot-path services are not competing for resources with less time-sensitive components such as Merchant Integration.

Components Identified

1. Merchant Integration Service
2. Wallet / Account Aggregation Service
3. Point Conversion Service
4. Fraud Detection & Voucher Security Service
5. Voucher / Gift Card Generation Service